

Part I: Theory Questions

(Chapter-wise Collection of All Theory Concepts)

Module 1: Introduction to Civil Engineering

1. Discuss scope of civil engineering. (W-24)
2. Define the civil engineering and enlist the branches of civil engineering. (S-25)
3. Discuss Branches of Civil Engineering. (W-18, W-19)
4. As a Civil Engineer discuss your role in society. (W-18)
5. Give objectives of riverfront development project and Sardar sarovar project. (S-25)
6. Explain role of civil engineer in infrastructure development. (W-22)
7. Enlist various fields of civil engineering and explain any two in brief. (S-20)
8. Write a short note on: Impact of infrastructural development on the economy of a country. (W-21)

Module 2: Civil Engineering Materials

1. Discuss various grades of concrete. (W-24)
2. What is FRP? Write uses and characteristics of FRP. (W-24)
3. Discuss classification of bricks and what are the properties of a good brick? (W-24, W-18)
4. How would you select the materials for low-cost housing? (W-24)
5. Enlist the various types of cement and discuss the properties with its uses. (S-25)
6. Enlist commonly used smart materials and explain any one. (S-25)
7. Explain physical and chemical properties of cement. (W-18)
8. Give Dimension of Modular Brick and discuss characteristics of first class brick. (W-18)
9. Write a short note on: Timber. (W-19)
10. Explain the ingredients of concrete. (S-20)
11. Discuss properties and uses of steel. (W-22)
12. Enlist various types of glasses and explain any two. (S-23)

Module 3: Building Planning and Construction

1. Discuss classification of buildings based on structure. **(W-24)**
2. Classify the buildings based on the structure. **(S-25)**
3. Discuss aims and objectives of town planning. **(W-24)**
4. Differentiate between English bond and Flemish bond. **(W-24)**
5. What are the elementary principles of building planning? Describe them briefly. **(W-24, W-18, S-25)**
6. Explain site plan and key plan. **(W-24)**
7. Describe the various principles of zoning. **(W-24)**
8. Enlist principles of planning and describe any one in detail. **(S-25, W-18)**
9. Draw cross section through window for a single wall for a residential building from foundation bottom to parapet top showing spread footing foundation for 30 cm thick wall. Label the components and provide dimensions. **(S-25)**
10. Draw the symbol of concrete, brick, wood, revolving door. And Write the abbreviation of PL, FSI, RCC. **(S-25)**
11. List the types of foundations. Explain various shallow foundations with sketch. **(S-25, W-18)**
12. Enlist different bonds of Brick and Describe English bond in detail. **(W-18)**
13. Explain Energy efficient Building. **(W-18)**
14. What is nominal dimensions for door and window? **(BCE Papers)**
15. Define: Substructure, Superstructure, Plinth, Parapet. **(S-20)**
16. Explain: Aspect, Prospect, Privacy, Roominess. **(W-21)**
17. Write a short note on: Load bearing structure vs Framed structure. **(W-22)**
18. Enlist various components of building and explain function of any two. **(S-23)**

Module 4: Surveying and Geomatics

1. Differentiate between plane surveying and geodetic surveying. **(W-24, W-18)**
2. Write applications of surveying. **(W-24)**
3. What do you understand by indirect ranging? Describe in detail with neat sketch. **(W-24)**
4. Discuss characteristics of contours with suitable sketches. **(W-24, W-18, BCE Papers)**
5. Explain temporary adjustments of level. **(W-24)**
6. Describe briefly the application and uses of GPS. **(W-24, S-25)**
7. Give difference between Surveyors and prismatic compass. **(S-25, W-18)**

8. Draw the conventional symbols for Survey station, Temple, Lake, and Double railway line. **(S-25)**
9. Enlist the various instruments used in chain survey with sketch. **(S-25, W-18)**
10. What is Total station? State its field applications. **(S-25, W-18)**
11. Draw sketch of Dumpy level and show all parts. **(W-18, BCE Papers)**
12. Give application of Remote sensing. **(W-18)**
13. Explain: Fundamental axes of Theodolite. **(S-19)**
14. Define: Bench mark, Reduced Level, Back sight, Fore sight. **(S-20)**
15. Explain reciprocal ranging. **(W-21)**
16. List classification of surveying based on instruments used. **(W-22)**
17. Explain characteristics of contours. **(S-23)**

Module 5: Water Resources and Environmental Engineering

1. What are the Indian standards for drinking water? **(W-24)**
2. Write drinking water quality standards as per BIS. **(S-25, W-18)**
3. What are the effects of solid waste? **(W-24)**
4. Discuss various pipes used in the house drainage plumbing system. **(W-24)**
5. Draw a typical plan for drainage of a small house. **(W-24)**
6. Differentiate between BOD and COD. **(S-25)**
7. Enlist various building services and objectives of building services. **(S-25)**
8. Define solid waste management. Give objectives and activities of solid waste management. **(S-25)**
9. Give classification of solid waste and describe why its segregation in wet and dry is necessary? **(W-18, BCE Papers)**
10. Describe house water connection with sketch. **(W-18)**
11. Write a short note on rain water harvesting systems. **(S-25, BCE Papers)**
12. Explain hydrological cycle with neat sketch. **(W-19)**
13. Explain methods of water conservation. **(S-20)**
14. Write a short note on: Global warming. **(W-22)**

Module 6: Transportation Engineering

1. Write advantages and disadvantages of BRTS. **(W-24)**
2. Describe pillars of smart city. **(W-24)**
3. Write short note on BRTS and Metro. **(S-25, BCE Papers)**
4. Describe important features of green building. **(S-25)**
5. Explain features of smart cities. **(W-18)**
6. Draw traffic signals for (i) speed limit (ii) No parking. **(W-18)**
7. Explain Aim, objectives and key features of BRTS. **(W-18)**
8. Explain concept and functions of smart cities. **(W-18)**
9. Compare BRTS and Metro rail. **(BCE Papers)**
10. Discuss the role of transportation in the development of a country. **(S-19)**
11. Enlist various modes of transportation and compare any two. **(W-21)**
12. Write a short note on: Intelligent Transport System (ITS). **(S-23)**

Part II: Numerical Problems

(Chapter-wise Collection of All Numerical Problems)

Module 1: Introduction to Civil Engineering (Numericals)

1. Convert (i) 10 m^2 to ft^2 (ii) 5 km to meter (iii) 1 Acre to m^2 (iv) 2 cubic meter to cu. Ft. (S-25)

Module 4: Surveying (Numericals)

1. Convert the following RB in to W.C.B: 1) $N55^\circ30'W$ 2) $S29^\circ00'W$ 3) $S23^\circ00'E$. (W-24)
2. The following are bearings of a closed traverse ABCDA. Find out interior angles and correct them if necessary. (W-24)

LINE	FB	BB
AB	$55^\circ30'$	$235^\circ00'$
BC	$113^\circ00'$	$293^\circ30'$
CD	$190^\circ00'$	$10^\circ00'$
DA	$279^\circ00'$	$99^\circ00'$

3. Determine the value of included angles in the closed traverse ABCD in clockwise direction for the given below bearings. Apply necessary check also. (S-25)

LINE	Fore Bearing
AB	40°
BC	70°
CD	210°
DA	280°

4. The following consecutive readings were taken with a level and a 4meter levelling staff on a continuously sloping ground: 0.570, 1.235, 1.750, 2.220, 2.665, 3.410, 1.005, 1.835, 2.165, 3.550, 0.825, 0.965, 1.730 and 2.320. The reduced level of the first point was 100 mt. Calculate reduced level of all stations using the rise and fall method and apply check. (S-25)
5. The following sequential readings were taken on a sloping ground. The first reading was taken on a BM of 50.000 m. Find out RL of each point and apply arithmetical check. 0.450, 0.560, 0.785, 1.200, 2.500, 3.480, 0.320, 0.655, 0.985, 1.300. (W-18)
6. The following bearings were taken on a closed traverse ABCD find out internal angles and correct them if necessary. (W-18)

Line	FB	BB
AB	$55^\circ30'$	$235^\circ00'$
BC	$113^\circ00'$	$293^\circ30'$
CD	$190^\circ00'$	$10^\circ00'$
DA	$279^\circ00'$	$99^\circ00'$

7. Convert the following Whole Circle Bearings to Reduced Bearings: (i) $65^{\circ}30'$ (ii) 140° (iii) $255^{\circ}45'$ (iv) $310^{\circ}30'$. **(W-19)**
8. The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth readings: 2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684 meters. Enter the above readings in a page of a level book and calculate the R.L. of points if the first reading was taken with a staff held on a bench mark of 432.384m. **(S-20)**
9. A series of offsets were taken from a chain line to an irregular boundary line at an interval of 10 m in the following order: 0, 2.50, 3.50, 5.00, 4.60, 3.20, 0 m. Calculate the area between the chain line and the irregular boundary by Trapezoidal rule and Simpson's rule. **(W-21)**



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